Name:- Abdullah Nasir.
Roll No:- 30.
Program:- BS CS.
Section:- (A).
Subject:- Programming fundamentals.
Teacher:- Sir Asad Majeed

Lab-2 Mannual.

TASK - 1.

Run the given programs and obeserve their Outputs.

(a)-

```
# include <iostream>
using namespace std;
int main () {
    char letter;
letter = 'A';
cout << letter << "\n";
letter = 'B';
cout << letter << "\n";
cin >> letter;
return 0;
}.
```

Output:-

A
B.

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(b)-

```
# include <iostream>.
using namespace std;
int main () {
    char letter;
    letter = 65;
  cout << letter << " \n";
    letter = 66;
  cout << letter << " \n";
     cin >> Letter;
     return 0;
}
```

Output:-

A (ASCII code of 65).
B (ASCII code of 66).

(c).

```
# include <iostream>
using namespace std;
int main ( ) {
   int letter;
   letter = 'A';
 cout << letter << "\n";
   letter = 'B';
 cout << letter << "\n";
   cin >> letter;
   retur 0;
}
```

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Output:-

65 (ASCII code of A.)
66 (ASCII code of B.)

TASK – 2.

Take two integers as input apply arthematic operations on them (+, -, *, /) as print them on screen.

```
# include < iostream>
using namespace std;
int main ( ) {
    int a = 10;
    int b = 2;
cout << a+b << endl;
cout << a-b << endl;
cout << a * b << endl;
cout << a / b << endl;
}
```

Output:-

12
8
20
5

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TASK – 3.

Write the output of the following code

```
#include <iostream>
using namespace std;
int main () {
    int val 1;
    int val 2;
    int val 3;
cout << "Please enter your 3 numbers:";
cin >> val 1 >> val 2 >> val 3;
cout << val 1 << "/n" << val 2 << "/n" <<
                                val 3 << "/n";
cout << val 3 << "/n" << val 2 << "/n" <<
                                val 1 << "/n";
    return 0;
}
```

Output :-

Please enter your 3 numbers are 1, 3, 8.

```
1     8
3     3
8     1
```

TASK – 4.

Write a C++ program to calculate the distance between the two points.

Note:- x_1, y_1, x_2, y_2 are all double values.

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```
#include <iostream>
#include <cMath>
using namespace std;
  int main () {
    double x1, y1, x2, y2 ;
       double distance;
    cout << "Enter x1:";
     cin >> x1 ;
    cout << "Enter y1:";
     cin>> y1 ;
    cout <<" Enter x2:";
     cin>> x2;
    cout <<" Enter y2:";
     cin>> y2 ;
    distance = sqrt (Pow (x2-x1 ,2) + Pow(y2-y2
                                    ,2));
     cout <<"Distance between two points
    is :" << distance << endl;
        return 0;
     }
```

Output:-

```
Enter x1 ; 1
Enter y1 : 2
Enter x2 : 4
Enter y2 : 6
```

Distance between two points is 5.

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TASK-5.

- Calculate the area of circle (area = pie $\times r^2$).

```
# include <iostream>
using namespace std;
int main ( ) {
    double r , area;
    const double pie = 3.14159 ;
    Cout <<" Enter radius of circle:" ;
     cin >> r ;
    area = pie * r * r ;
    cout << " Area of a circle :" << area <<
                                        end l;
     return 0;
   }
```

Output:-

```
Enter radius of circle : 5.
Area of a circle : 78.5397.
```

- Calculate the area of rectangle (area = l x w)

```
# include <iostream>
using namespace std;
int main ( ) {
     float length , width , area;
    cout << "Enter length of rectangle :" ;
     cin>> length ;
```

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```
cout << "Enter width of rectangle :";
  cin >> width ;
  area = length * width;
  cout << Area of rectangle :" << area << end l;
     return 0;
     }
```

Output:-

```
Enter length of rectangle : 4.2
Enter width of rectangle : 5.1
area of rectangle : 21.42
```

- Calculate the Area of triangle (area = b x h x 0.5)

```
# include <iostream>
using namespace std;
int main () {
    float base , height , area ;
    cout << "Enter base of triangle: ";
     cin >> base ;
    Cout << "Enter height of triangle:" ;
     cin >> height ;
    area = base x height x 0.5
   Cout << "Area of triangle:" << area <<
                                  end l;
      return 0;
   }
```

Output:-

```
Enter base of triangle : 5.6
Enter height of triangle : 8.1
Area of triangle : 22.68.
```

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TASK - 6.

Total purchase A customer in a store is purchasing five items, The Prize of items are.

Prize of item 1: $ 12.95.
Prize of item 2: $ 24.95.
Prize of item 3: $ 6.95.
Prize of item 4: $ 14.95.
Prize of item 5: $ 3.95.

write a program that hold the prize five items in five variables, Display each item prize subtotal of the scale, the amount of scales tax and the total Assumes of scales tax; 6%.

```
# include <iostream>
using namespace std;
 int main () {
    float item 1 = 12.95 f ;
    float item 2 = 24.95 f;
    float item 3 = 6.95 f;
    float item 4 = 14.95 f;
    float item 5 = 3.95 f;
float subtotal , sales tax , total ;
 const float Tax-rate = 0.06 f ;
Subtotal = item 1 + item 2 + item 3 + item 4 +
                                    item 5;
  Sales Tax = subtotal + Tax-Rate ;
   Total = Subtotal + Sales Tax ;
```

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```
Cout << "Price of item 1: $" << item1 << endl;
cout << "Price of item 2: $" << item2 << endl;
cout << "Price of item 3: $" << item3 << endl;
cout << "Price of item 4: $" << item4 << endl;
cout << "Price of item 5: $" << item5 << endl;
cout << "Subtotal of the sale: $" << subTotal <<
cout << "Amount of saleTax (6%): $" << saletax <<
                                                 endl;
cout << "Total: $" << Total << endl;              endl;
    return 0;
}
```

Output:-

```
Price of item 1 : $ 12.95
Price of item 2 : $ 24.95
Price of item 3 : $ 6.95
Price of item 4 : $ 14.95
Price of item 5 : $ 3.95
Subtotal of the sale : 63.75
Amount of Sales Tax : 3.825
Total : $ 67.575
```

Bonus Task.

Write a program that inputs a five-digit integers, seprate the integer into its individual digits and prints the digits seprated from one another by three spaces each.

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```
# include <iostream>
using namespace std;
int main ( ) {
    cout << "Enter a five digit Integer:" ;
    cin >> num;
    int d1 = num / 10000 ;
    int d2 = (num / 1000) % 10;
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = (num % 10) ;
cout << d1 << " " << d2 << " " << d3 << " "
        << d4 << " " << d5 << " " << endl;
    return 0;
}
```

Home Task.

Task-1:-

```
# include <iostream>
using namespace std;
int main () {
    int var 1 ; Var 2;
    cin >> Var 1 >> Var 2;
    Swap ( var 1 , Var 2 );
    cout << "Var 1 =" << Var 1 << "Var 2 =" Var 2;
}
```

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Task-2:-

```
# include <iostream>
using namespace std;
int main () {
  char ch;
  cin >> ch;
  cout << char (toupper (ch));
  }
```

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Task 1 (a)

```cpp

#include <iostream>
using namespace std;

int main() {
    char letter;

    letter = 'A';
    cout << letter << "\n";

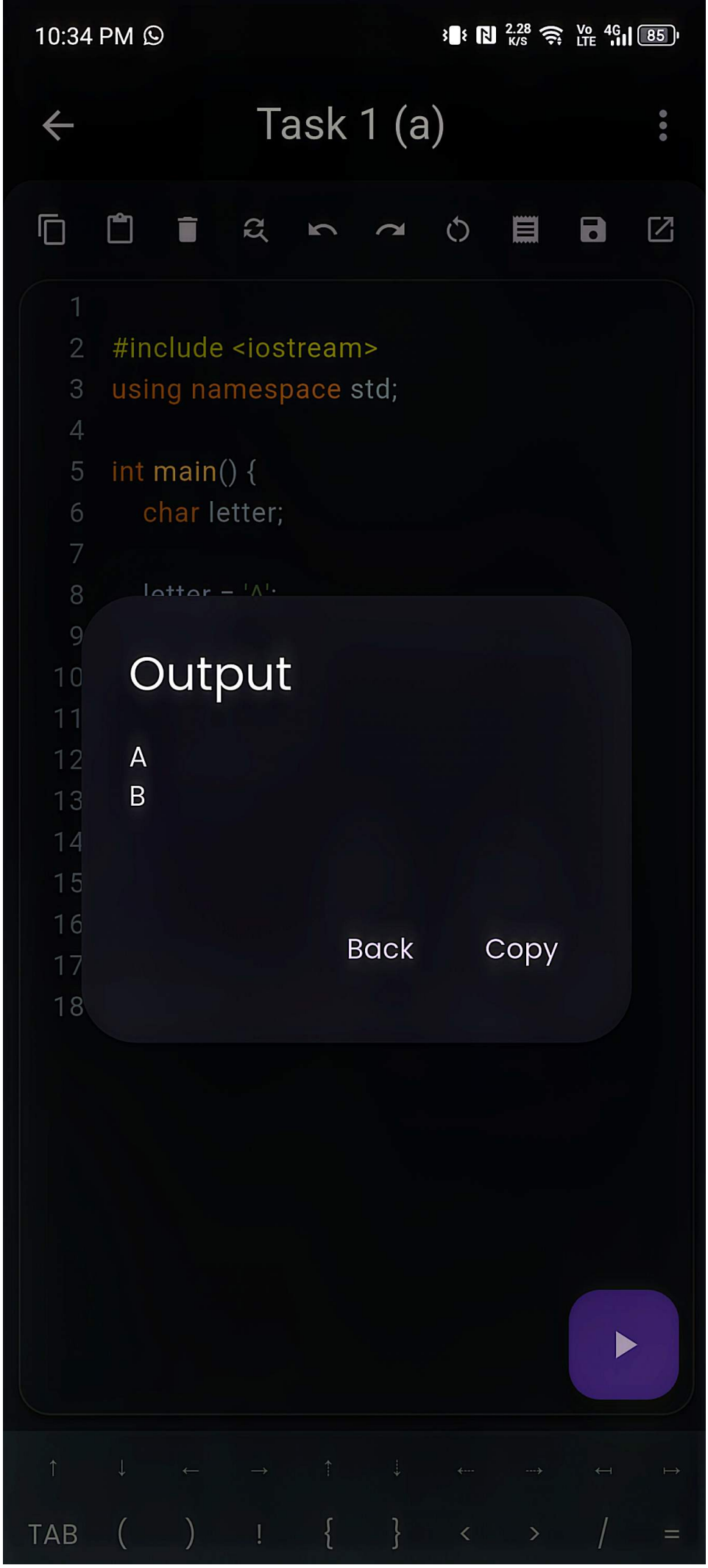
    letter = 'B';
    cout << letter << "\n";

    cin >> letter;

    return 0;
}

```

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10:34 PM
Task 1 (a)
#include <iostream>
using namespace std;
int main() {
char letter;
Output
A
B
Back
Copy

10:35 PM

Task 1(b)

```cpp
#include <iostream>
using namespace std;

int main() {
    char letter;

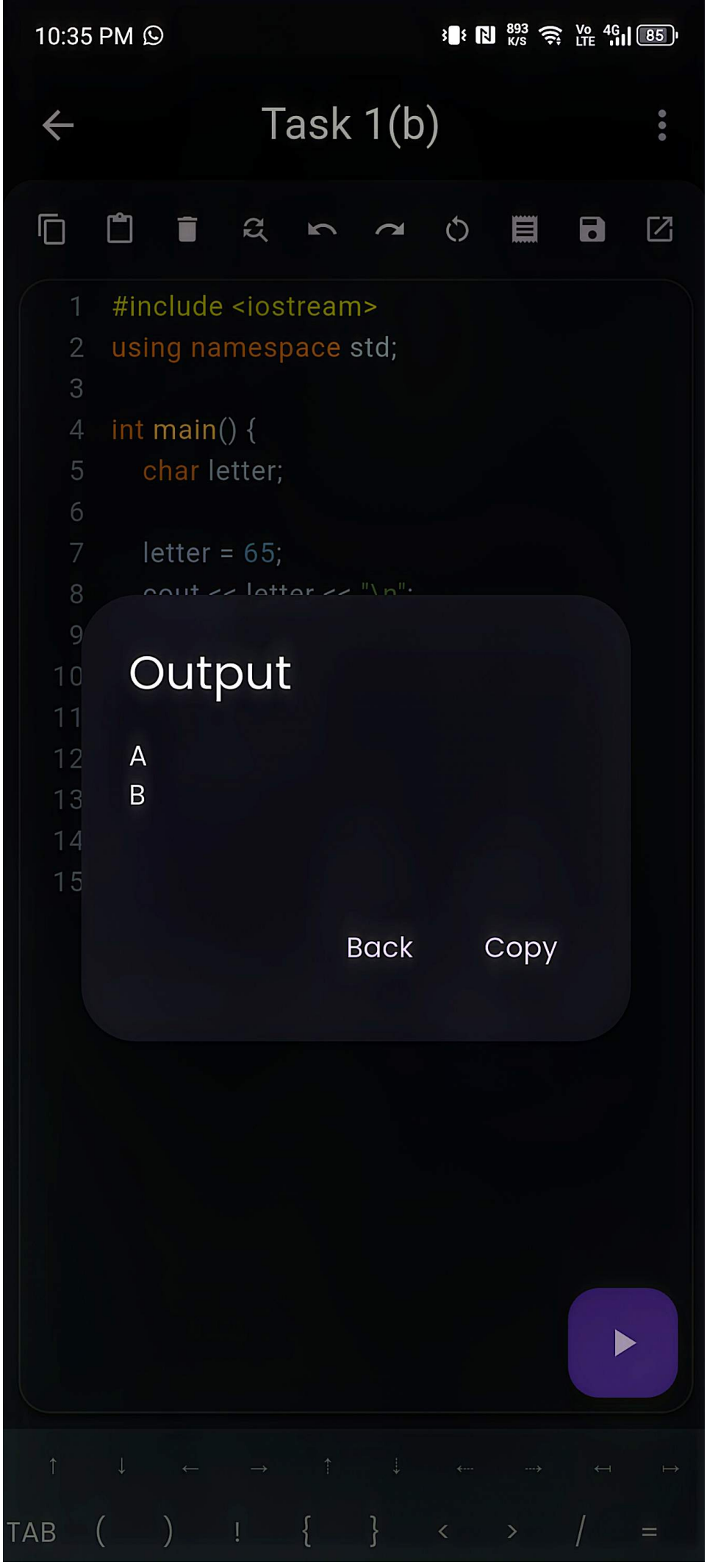
    letter = 65;
    cout << letter << "\n";

    letter = 66;
    cout << letter << "\n";

    cin >> letter;
    return 0;
}
```

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Task 1(b)
#include <iostream>
using namespace std;
int main() {
char letter;
letter = 65;
Output
A
B
Back
Copy
TAB ( ) ! { } < > / =

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Task 1(c)

```cpp
#include <iostream>
using namespace std;

int main() {
    int letter;

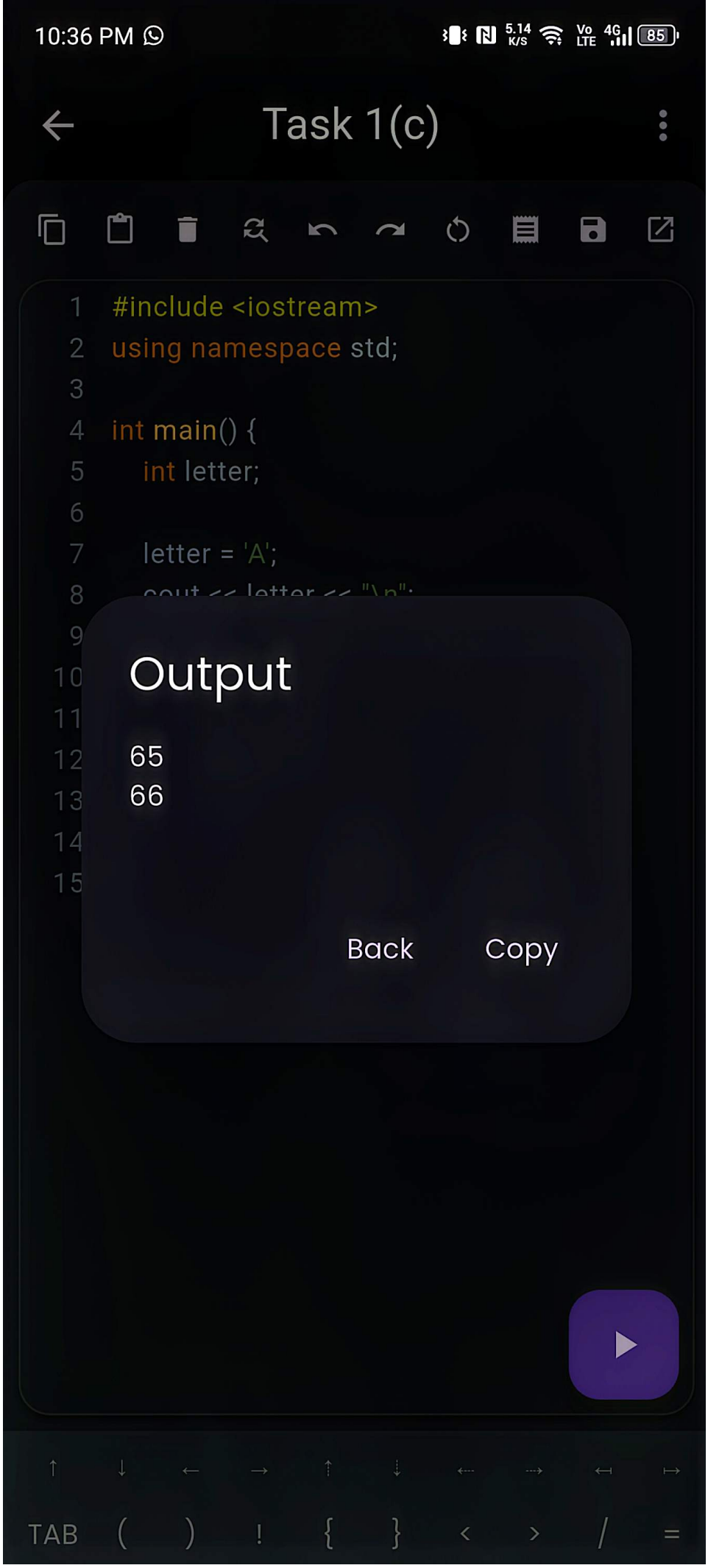
    letter = 'A';
    cout << letter << "\n";

    letter = 'B';
    cout << letter << "\n";

    cin >> letter;
    return 0;
}
```

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Task 1(c)
#include <iostream>
using namespace std;
int main() {
int letter;
letter = 'A';
Output
65
66
Back
Copy
TAB ( ) ! { } < > / =

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Task 2

```cpp
#include <iostream>
using namespace std;

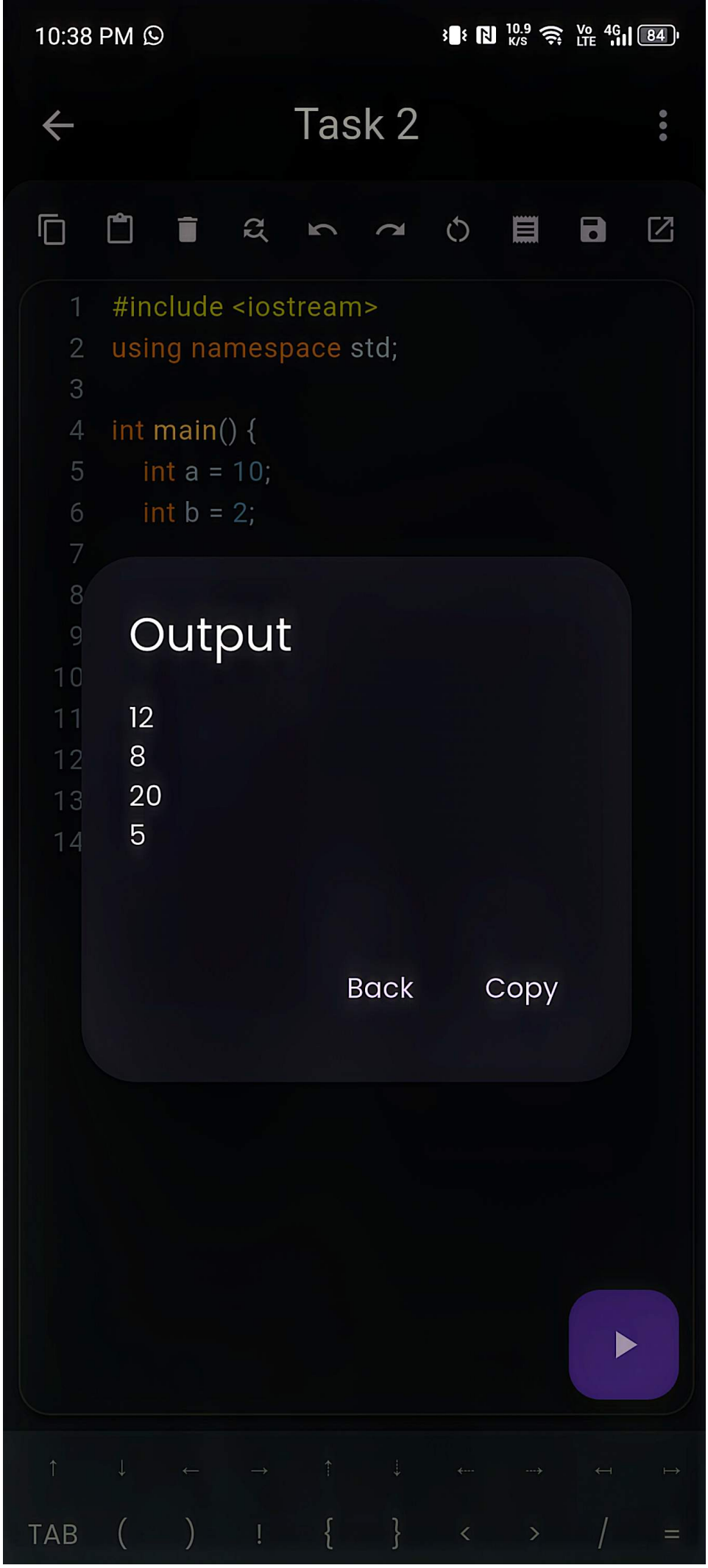
int main() {
    int a = 10;
    int b = 2;

    cout << a + b << endl;
    cout << a - b << endl;
    cout << a * b << endl;
    cout << a / b << endl;

    return 0;
}
```

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Task 2
#include <iostream>
using namespace std;
int main() {
int a = 10;
int b = 2;
Output
12
8
20
5
Back
Copy
TAB ( ) ! { } < > / =

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Task 3

```cpp
#include <iostream>
using namespace std;

int main() {
    int val1, val2, val3;

    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

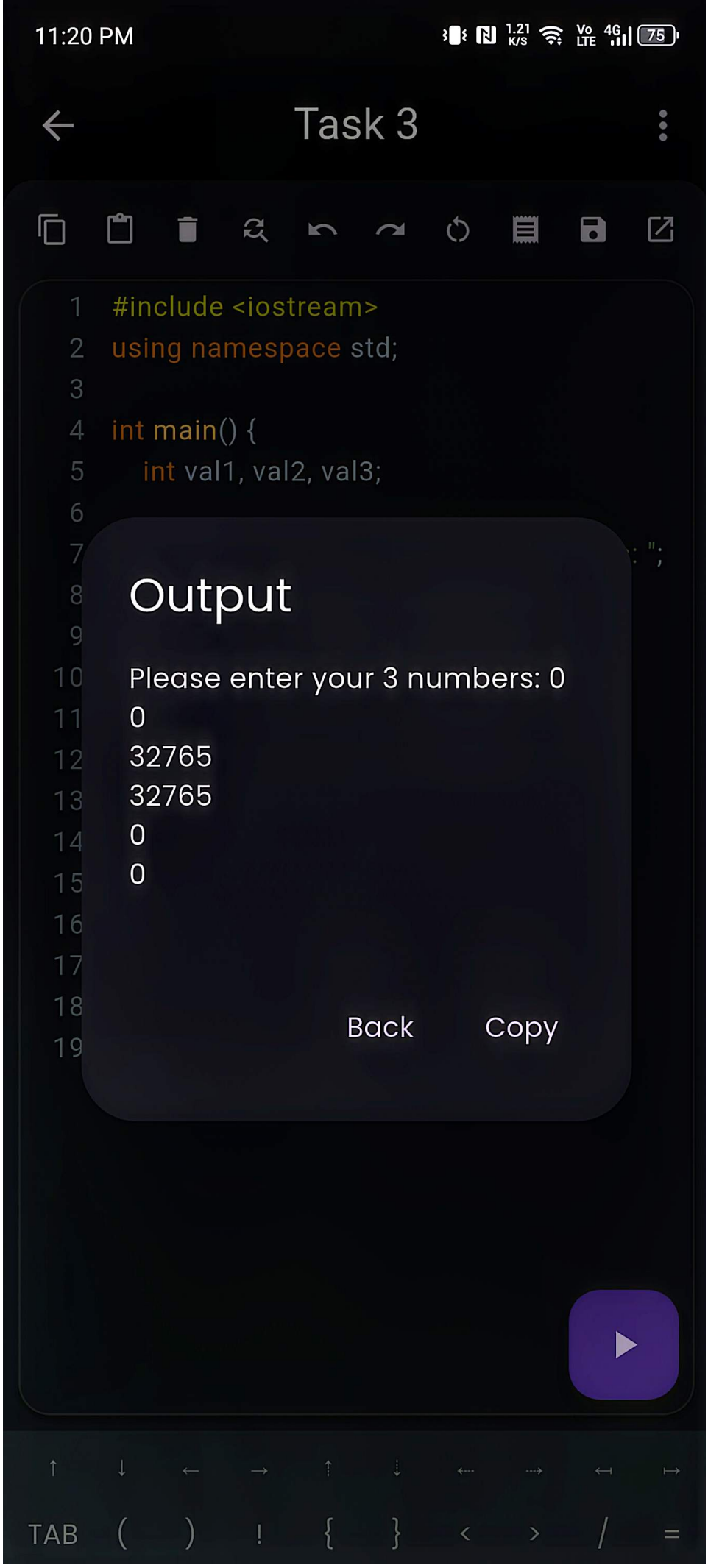
    cout << val1 << endl;
    cout << val2 << endl;
    cout << val3 << endl;

    cout << val3 << endl;
    cout << val2 << endl;
    cout << val1 << endl;

    return 0;
}
```

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Task 3
#include <iostream>
using namespace std;
int main() {
int val1, val2, val3;
Output
Please enter your 3 numbers: 0
0
32765
32765
0
0
Back
Copy

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11:21 PM

Task 4

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    double x1, y1, x2, y2;
    double distance;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

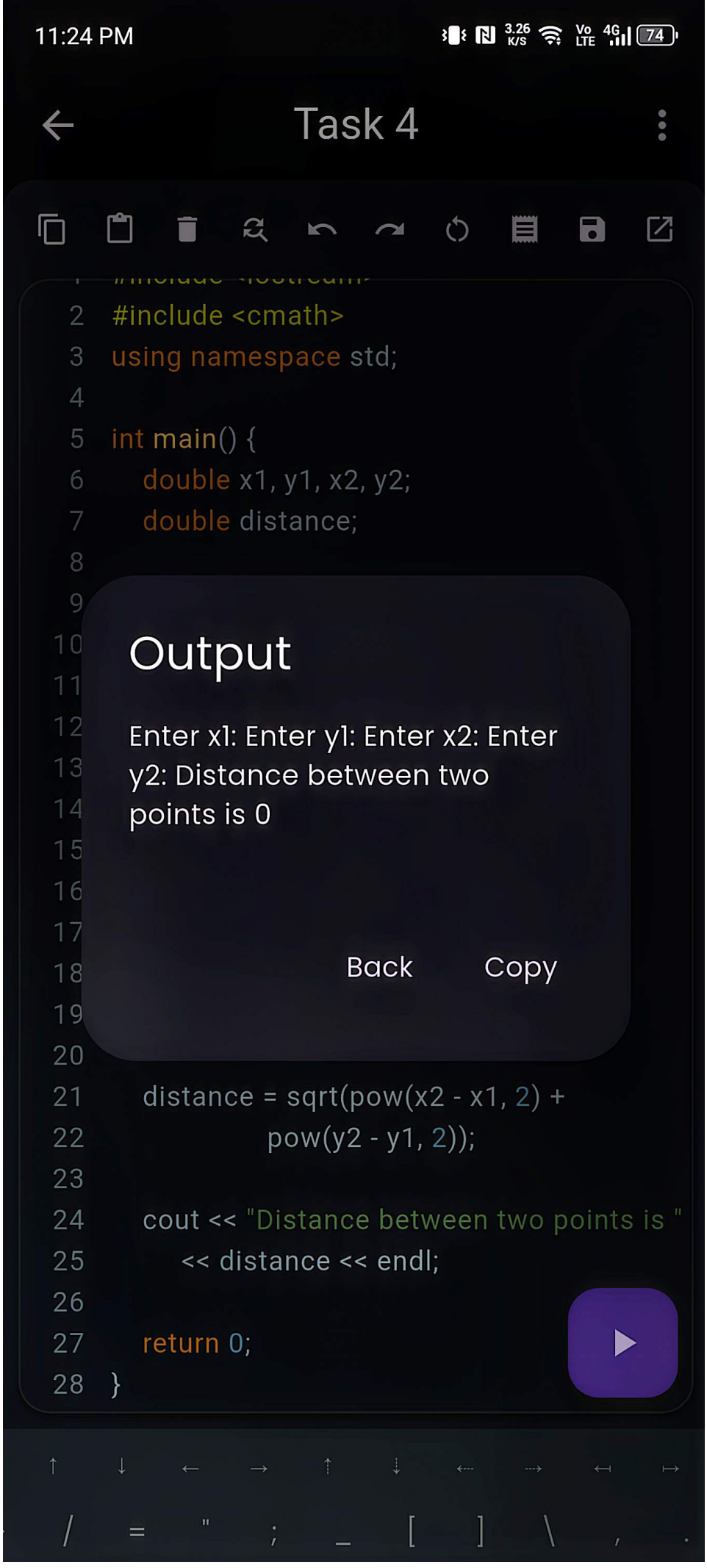
    cout << "Enter y2: ";
    cin >> y2;

    distance = sqrt(pow(x2 - x1, 2) +
                pow(y2 - y1, 2));

    cout << "Distance between two points is "
        << distance << endl;

    return 0;
}
```

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11:24 PM
Task 4
#include <cmath>
using namespace std;
int main() {
double x1, y1, x2, y2;
double distance;
Output
Enter x1: Enter y1: Enter x2: Enter y2: Distance between two points is 0
Back
Copy
distance = sqrt(pow(x2 - x1, 2) +
pow(y2 - y1, 2));
cout << "Distance between two points is "
<< distance << endl;
return 0;
}

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Task 5(a)

```cpp
#include <iostream>
using namespace std;

int main() {
    double r, area;
    const double pie = 3.14159;

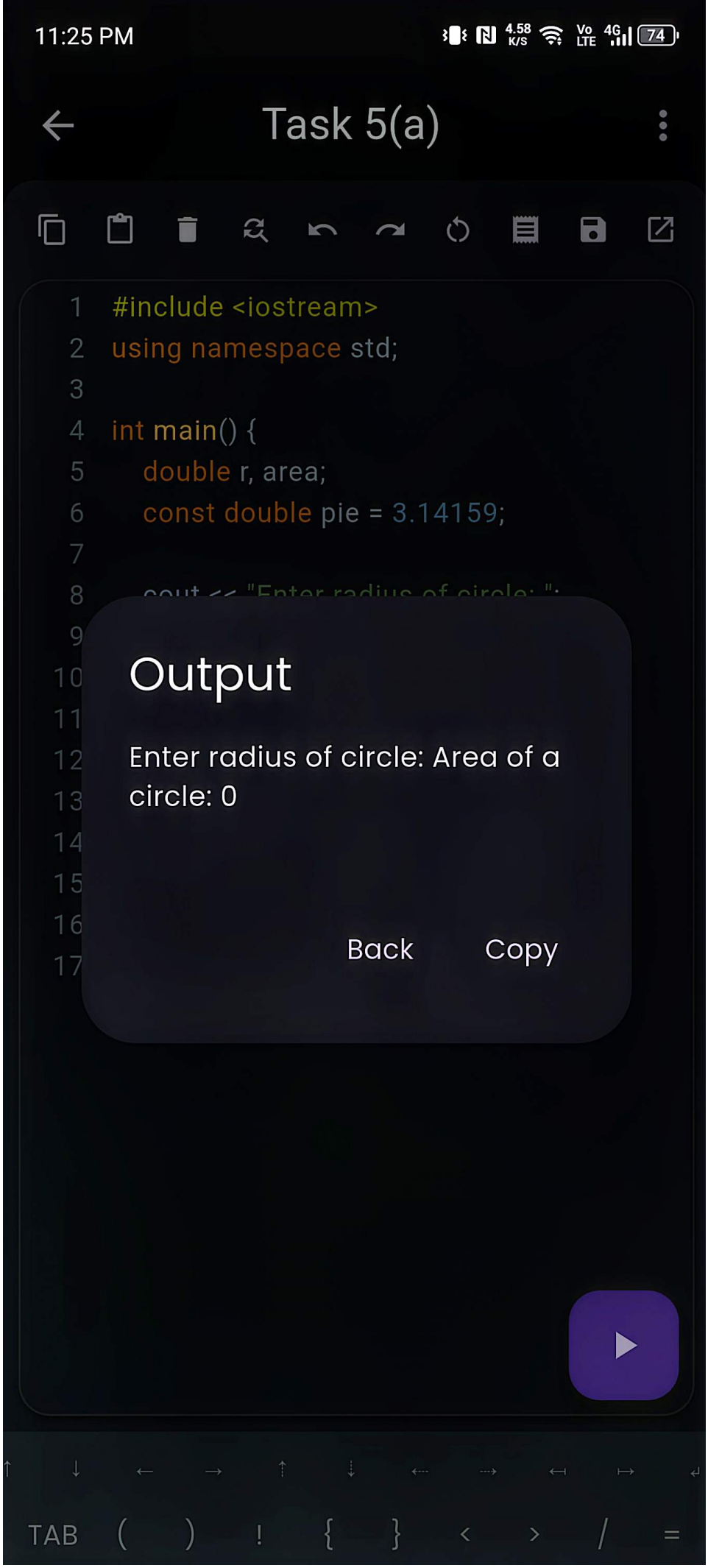
    cout << "Enter radius of circle: ";
    cin >> r;

    area = pie * r * r;

    cout << "Area of a circle: "
         << area << endl;

    return 0;
}
```

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Task 5(a)
#include <iostream>
using namespace std;
int main() {
double r, area;
const double pie = 3.14159;
Output
Enter radius of circle: Area of a circle: 0
Back
Copy
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Task 5 (b)

```cpp
#include <iostream>
using namespace std;

int main() {
    float length, width, area;

    cout << "Enter length of rectangle: ";
    cin >> length;

    cout << "Enter width of rectangle: ";
    cin >> width;

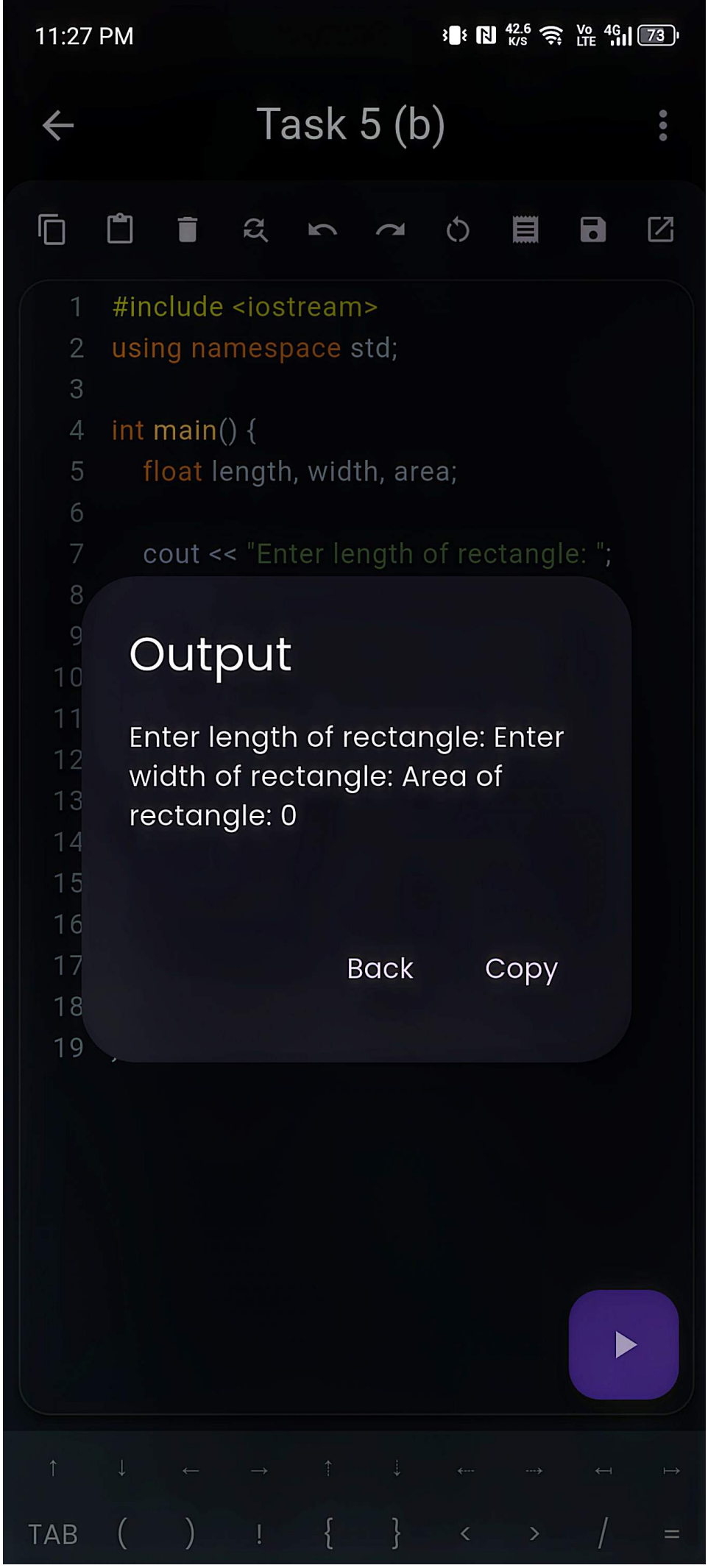
    area = length * width;

    cout << "Area of rectangle: "
         << area << endl;

    return 0;
}
```

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Task 5 (b)
#include <iostream>
using namespace std;
int main() {
float length, width, area;
cout << "Enter length of rectangle: ";
Output
Enter length of rectangle: Enter width of rectangle: Area of rectangle: 0
Back
Copy
TAB ( ) ! { } < > / =

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Task 5(c)

```cpp
#include <iostream>
using namespace std;

int main() {
    float base, height, area;

    cout << "Enter base of triangle: ";
    cin >> base;

    cout << "Enter height of triangle: ";
    cin >> height;

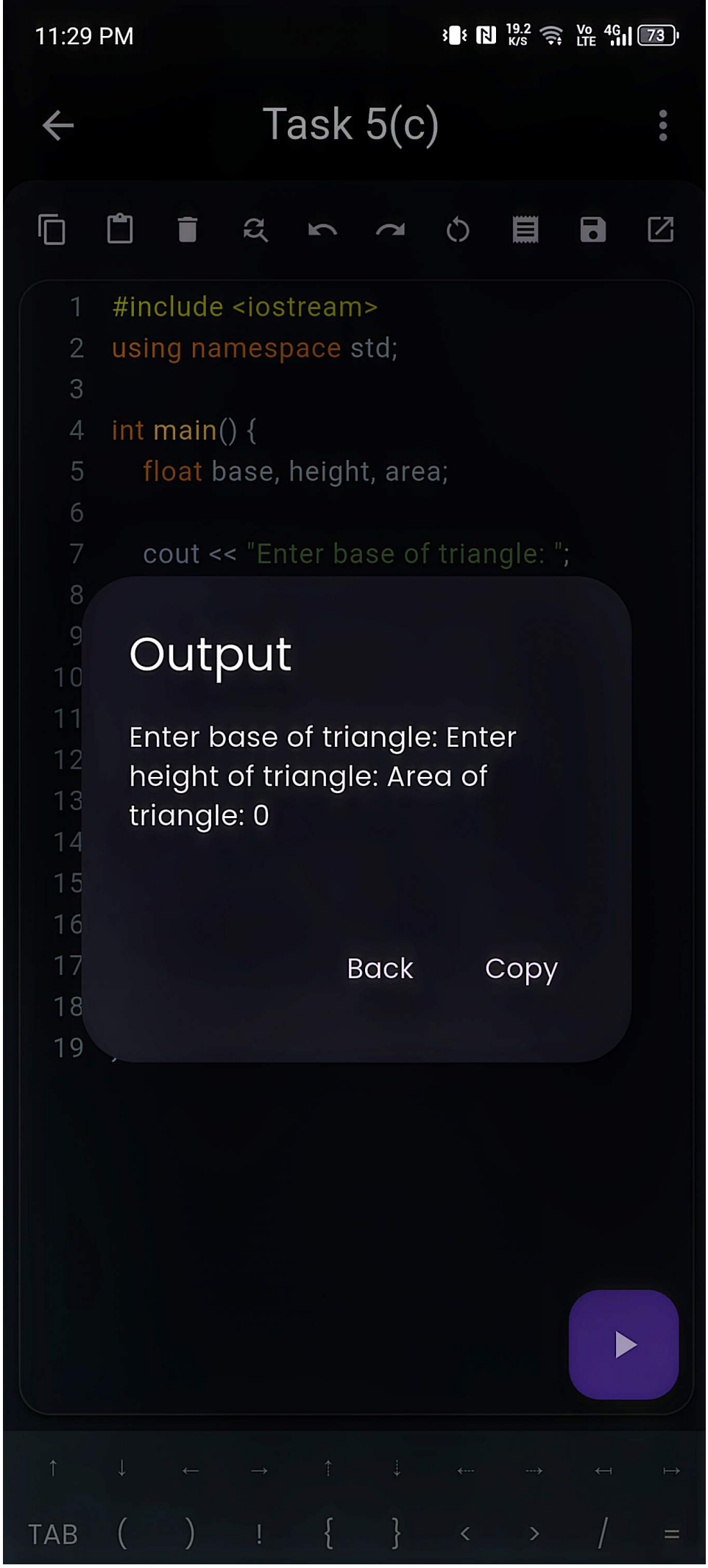
    area = base * height * 0.5;

    cout << "Area of triangle: "
         << area << endl;

    return 0;
}
```

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Task 5(c)
#include <iostream>
using namespace std;
int main() {
float base, height, area;
cout << "Enter base of triangle: ";
Output
Enter base of triangle: Enter height of triangle: Area of triangle: 0
Back
Copy
TAB ( ) ! { } < > / =

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11:33 PM

Task 6

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    float item1 = 12.95;
    float item2 = 24.95;
    float item3 = 6.95;
    float item4 = 14.95;
    float item5 = 3.95;

    float subtotal, salesTax, total;
    const float taxRate = 0.06;

    subtotal = item1 + item2 + item3
            + item4 + item5;

    salesTax = subtotal * taxRate;
    total = subtotal + salesTax;

    cout << fixed << setprecision(2);

    cout << "Price of item 1: $" << item1 << enc
    cout << "Price of item 2: $" << item2 << enc
    cout << "Price of item 3: $" << item
    cout << "Price of item 4: $" << ite
    cout << "Price of item 5: $" << ite
```

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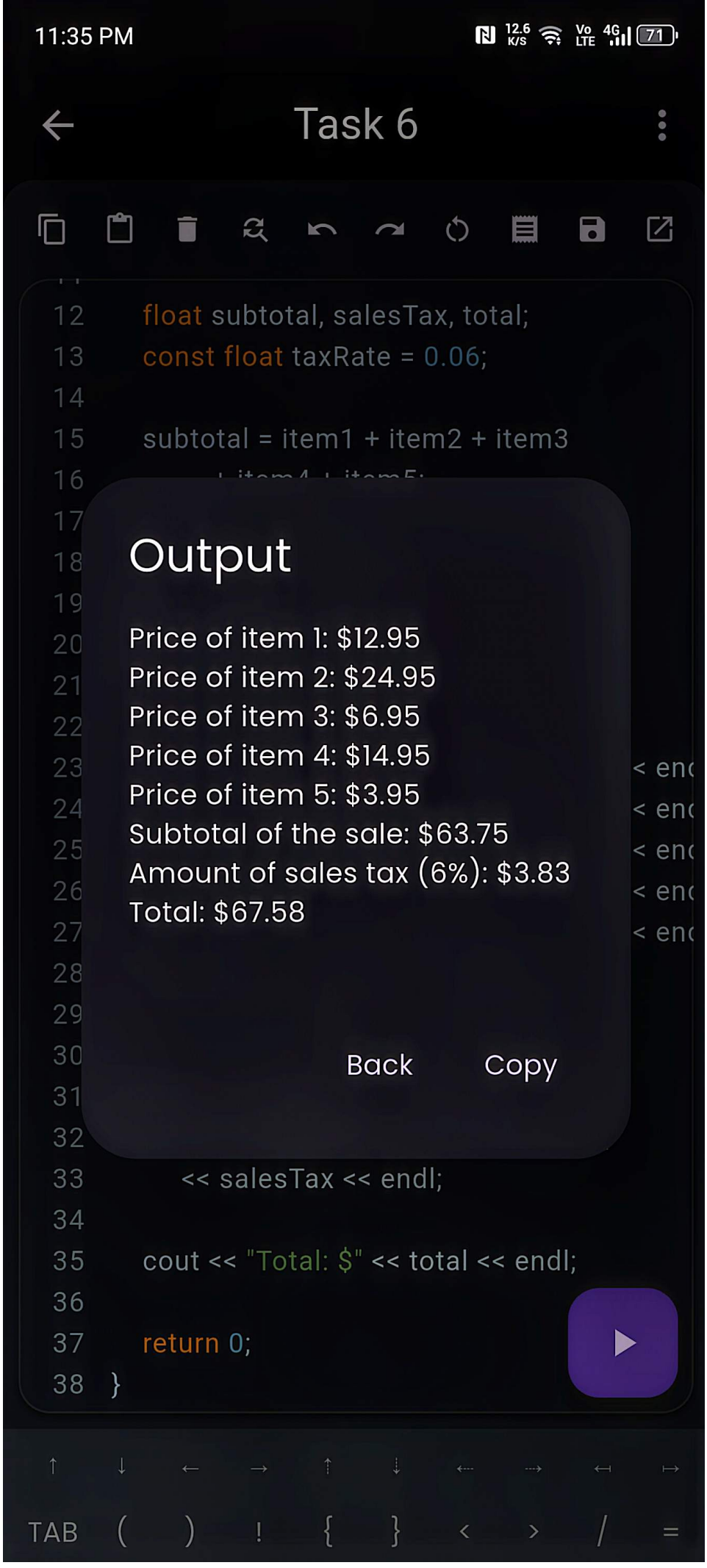
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Task 6

```
12    float subtotal, salesTax, total;
13    const float taxRate = 0.06;
14
15    subtotal = item1 + item2 + item3
16            + item4 + item5;
17
18    salesTax = subtotal * taxRate;
19    total = subtotal + salesTax;
20
21    cout << fixed << setprecision(2);
22
23    cout << "Price of item 1: $" << item1 << enc
24    cout << "Price of item 2: $" << item2 << enc
25    cout << "Price of item 3: $" << item3 << enc
26    cout << "Price of item 4: $" << item4 << enc
27    cout << "Price of item 5: $" << item5 << enc
28
29    cout << "Subtotal of the sale: $"
30        << subtotal << endl;
31
32    cout << "Amount of sales tax (6%): $"
33        << salesTax << endl;
34
35    cout << "Total: $" << total << endl;
36
37    return 0;
38 }
```

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Task 6
float subtotal, salesTax, total;
const float taxRate = 0.06;
subtotal = item1 + item2 + item3
Output
Price of item 1: $12.95
Price of item 2: $24.95
Price of item 3: $6.95
Price of item 4: $14.95
Price of item 5: $3.95
Subtotal of the sale: $63.75
Amount of sales tax (6%): $3.83
Total: $67.58
Back
Copy
<< salesTax << endl;
cout << "Total: $" << total << endl;
return 0;
}
TAB

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Bonus Task

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a five digit integer: ";
    cin >> num;

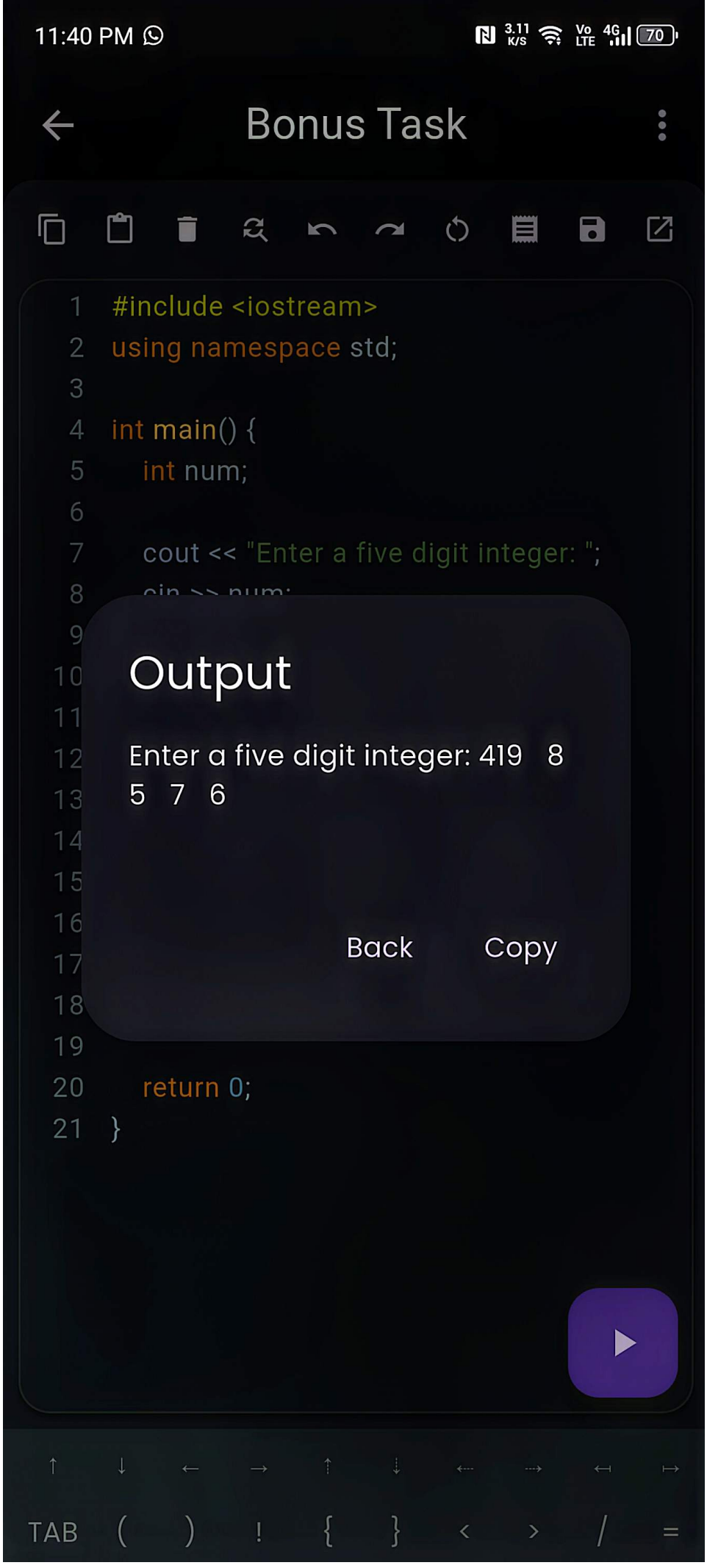
    int d1 = num / 10000;
    int d2 = (num / 1000) % 10;
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = num % 10;

    cout << d1 << "   " << d2 << "   "
         << d3 << "   " << d4 << "   "
         << d5 << endl;

    return 0;
}
```

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Bonus Task
#include <iostream>
using namespace std;
int main() {
int num;
cout << "Enter a five digit integer: ";
return 0;
}
Output
Enter a five digit integer: 419 8
5 7 6
Back
Copy
TAB ( ) ! { } < > / =

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Home Task 1

```cpp
#include <iostream>
using namespace std;

int main() {
    int var1, var2;

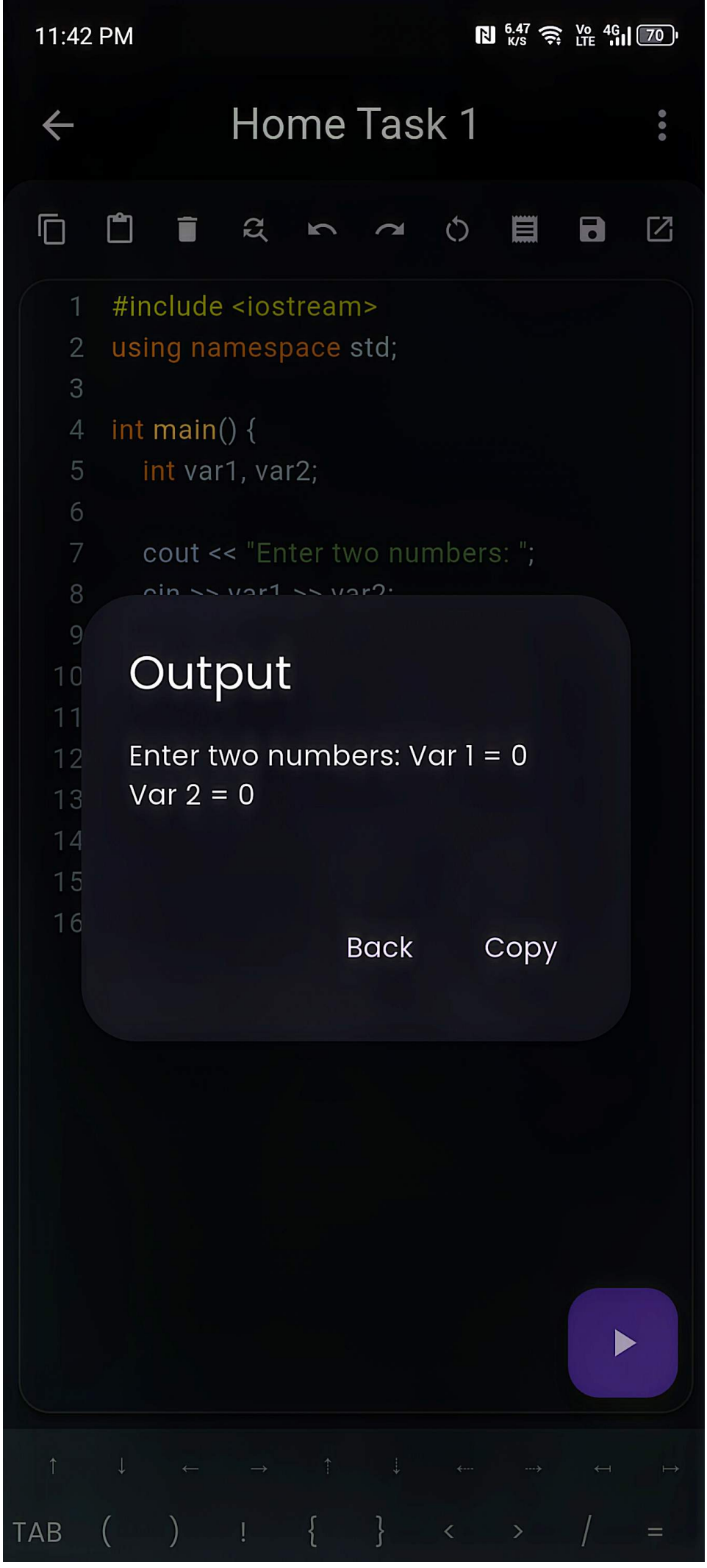
    cout << "Enter two numbers: ";
    cin >> var1 >> var2;

    swap(var1, var2);

    cout << "Var 1 = " << var1 << endl;
    cout << "Var 2 = " << var2 << endl;

    return 0;
}
```

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Home Task 1
#include <iostream>
using namespace std;
int main() {
int var1, var2;
cout << "Enter two numbers: ";
Output
Enter two numbers: Var 1 = 0
Var 2 = 0
Back
Copy
TAB

CamScanner

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Home Task 2

```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;

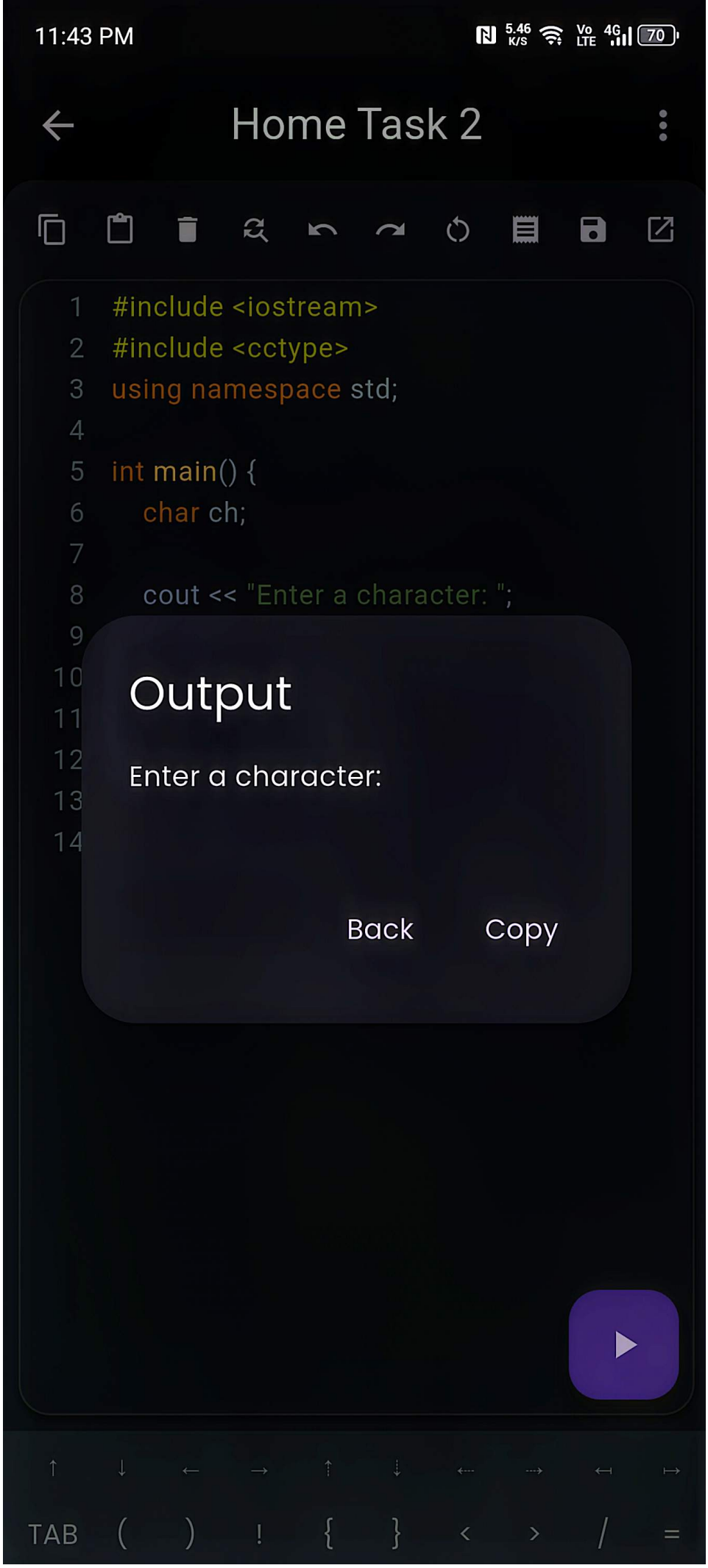
    cout << "Enter a character: ";
    cin >> ch;

    cout << char(toupper(ch)) << endl;

    return 0;
}
```

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Home Task 2
#include <iostream>
#include <cctype>
using namespace std;
int main() {
char ch;
cout << "Enter a character: ";
Output
Enter a character:
Back
Copy
TAB ( ) ! { } < > / =

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